Parsing Architecture and Standard Raw Event Format (SREF)

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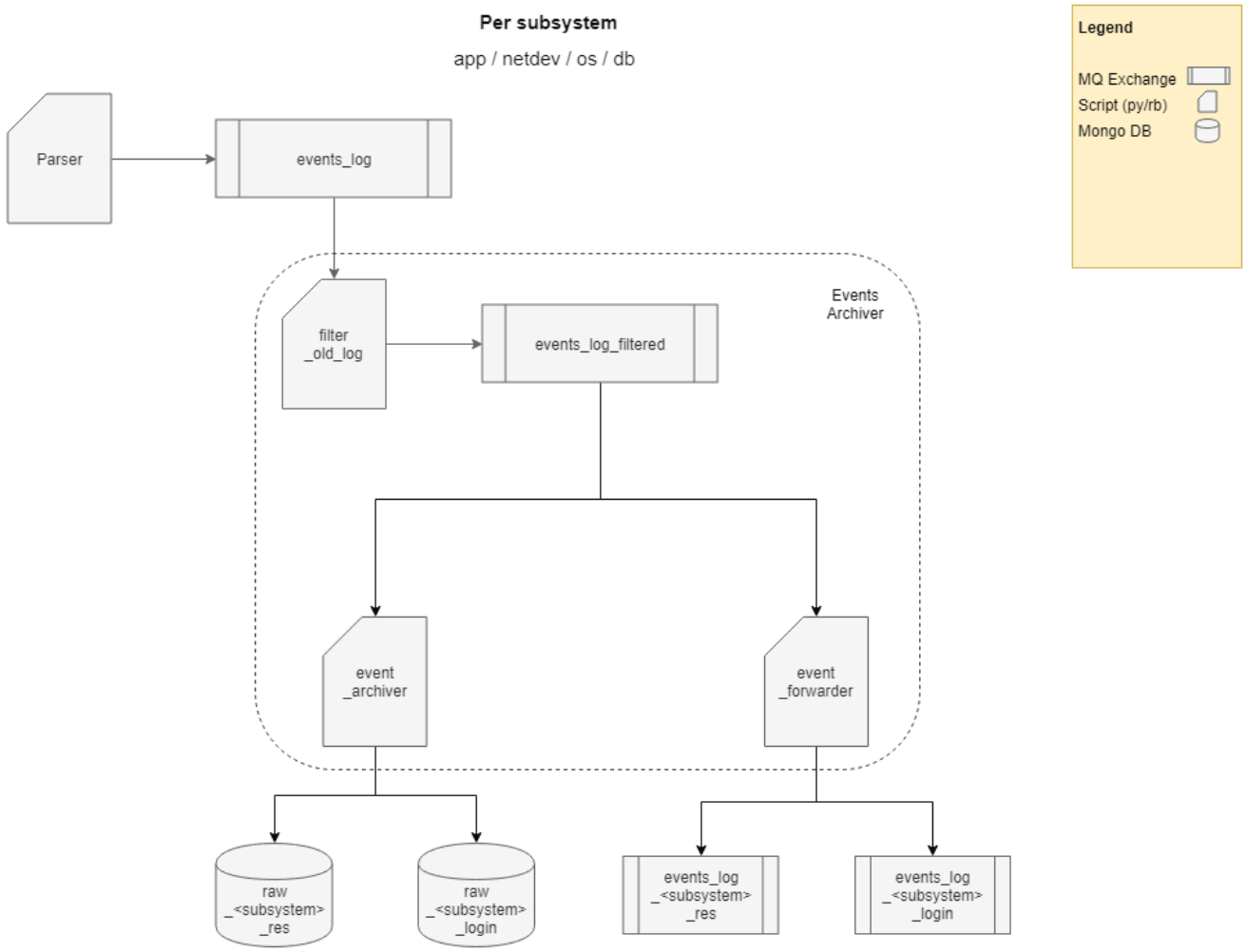
Overall Architecture

At the moment, only applies to DB, APP and NETDEV



Routing Key

Format

```
app_key.sub_system.action_category.res_type.action_result.action_type
```

Standard Raw Event Format (SREF)

Fields

Required Fields

- `time_received` : Epoch time when the log was processed in our analytic server
- `time_log` : Epoch time when the log was generated at the source
- `user` : User ID or name of the actor (NOTE: must be **lowercase**)
- `user_host` : System which hosts the `user` (IP or Hostname)
- `session_id` (optional): ID of the user session
- `app_type` : Type of application or system, e.g. os_windows, db_oracle, netdev_paloalto_firewall, sharepoint
- `app_host` : The system which hosts the app
  - `app_host.ip` : IP address
  - `app_host.hostname` (optional): Hostname
  - `app_host.role` (optional): Sensor agent role name
- `app_key` : The main identifier for `app_host` - this solves the issue of multiple `app_host` identifiers
  - `app_key.key` : Role / Hostname / IP (In order of preference, left-to-right)
  - `app_key.type` : Describes the type of key, can be `ip` , `hostname` or `role`
- `action_category` : Overall generic category of the event
  - `action_category = login` : For all login/logout events
  - `action_category = res` : Resource related events
- `action_type` : Describes what was done to the resource
  - Refer to the next section for more examples
- `action_result` : Whether the action was successful - either `success` or `failure`
- `res_type` : Type of resource that was affected
  - Refer to the next section for more examples
- `res_name` : Name of the resource that was affected

- `raw` : The original raw message generated by the event

Special Fields

These fields should be present for specific events, to assist algos for classification

- `action_extra` (optional): Resource + Action specific data
  - `action_extra.duration` : e.g. for DB row access, stores the query duration
  - `action_extra.is_privileged` : `True` if the action is privileged, e.g. user create/delete
  - `action_extra.is_service_account` : `True` if the the action was carried out system account or service account (an account that is non-interactive)
- `action_result_reason` (optional): Reason that the action failed
- `res_info` (optional): More info about the resource affected
  - e.g. if `action_type: perm.add` , res_info would contain the list of permissions
  - e.g. if `res_type: group` and `action_type: user.add` → `res_info` would be the user name
- `user_info` (optional): More info about the user that doesn't fit in the other fields

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Example

User AAA (using the IP 192.168.10.10) FAILS to grant the user BBB permission XXX, YYY, ZZZ on SharePoint (an application) which has the role name UATSP1. This occurred on 15 June 2020 at 3 pm (SGT), and we received the log 1 second later.

- `time_received` : `1592204401`
- `time_log` : `1592204400`
- `user` : `AAA`
- `user_host` : `192.168.10.10`
- `app_type` : `sharepoint`
- `app_host` : `{ ip: '192.168.9.9', role: 'UATSP1' }`
- `app_key` : `{ key: 'UATSP1', type: 'role' }`
- `action_category` : `res`
- `action_type` : `perm.add`
- `action_result` : `failure`
- `action_result_reason` : `no_permission`

- `action_extra`: `{ is_privileged: true }`
- `res_type`: `user`
- `res_name`: `BBB`
- `res_info`: `'XXX, YYY, ZZZ'`
- `raw`: `User AAA (using the IP 192.168.10.10) FAILS to grant the user BBB permission XXX, YYY, ZZZ on SharePoint (an application) which has the role name UATSP1.`

```
UATSP1.app.res.user.success.perm.add
```

Event Mapping

Login

- `action_category`: Every event will be `login`
- `res_type`: This will be the sub system - `db` / `os` / `netdev` / `app`
- `action_type`: `start` / `end`

Description	action_category	res_type	action_type
Login to a Database	login	db	start
Logout from Database	login	db	end
Login to Office 365	login	app	start
Login to Cisco Switch	login	netdev	start

Res

- `action_category`: Every event will be `res`
- `res_type`: The general name of the resource in lowercase
- `action_type`: Try to map actions to these
  - `create` / `read` / `update` / `delete`
  - `perm.add` / `perm.delete`
  - `execute` / `start` / `stop`

General

Description	action_category	res_type	action_type
Create user	res	user	create
Delete user	res	user	delete
Grant permissions to user	res	user	perm.add
Revoke permissions from user	res	user	perm.delete
Add user to group/role	res	group	user.add
Remove user from group/role	res	group	user.delete
Configuration changes	res	config	update
Audit settings change	res	audit	update

Database (specific)

Description	action_category	res_type	action_type
Start a database	res	database	start
Shutdown a database	res	database	stop
Create database	res	database	create
Drop database	res	database	delete
Create cluster	res	cluster	create
Drop cluster	res	cluste	delete
Create table	res	table	create
Drop table	res	table	delete
Modify table columns	res	table	update
Select	res	rows	read
Insert	res	rows	create
Update	res	rows	update
Delete	res	rows	delete